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AN ANALYSIS OF GENDER WAGE DIFFERENTIALS ACROSS SECTORS IN GEORGIA

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ABSTRACT

The persistence of gender-based wage disparities represents a critical challenge for both scholars and policymakers seeking to promote labor market equity. This study examines the determinants and structural characteristics of the gender wage gap in the Georgian labor market, emphasizing sectoral differentiation, male-dominated industries, and demographic composition. Utilizing secondary panel data from the Georgian National Statistics Bureau spanning the period 1999–2023, the analysis employs multiple regression and analysis of variance (ANOVA) techniques to assess the relationship between gender, sector, and average monthly earnings. The empirical results indicate that both time and gender exert a statistically significant influence on wage variation, with male employees consistently receiving higher remuneration than their female counterparts. Furthermore, the findings confirm that male-dominated sectors—such as manufacturing, wholesale and retail trade, and finance—exhibit markedly wider gender wage gaps, reflecting entrenched structural and institutional inequalities. Population dynamics further contribute to wage disparities, as evidenced by the positive association between the 45–65 age cohort and the magnitude of the wage gap. These outcomes corroborate international evidence linking occupational segregation, human capital disparities, and institutional discrimination to persistent gender-based wage inequality. The study contributes to the broader discourse on gender and labor economics by providing empirical insights into the Georgian context and underscores the necessity of policy interventions aimed at enhancing pay transparency, promoting gender mainstreaming, and strengthening formalized compensation systems to mitigate discriminatory wage practices.

INTRODUCTION

The gender wage gap has become the subject of wide-ranging academic research. According to Ahmed and Maitra (2015), the explicit aim of this body of work is to determine the extent to which the residual gender wage difference—after adjusting for differences in

benefactions—can be attributed to discrimination, as opposed to differences in productive attributes. This phenomenon is often referred to as the “discrimination effect.” From a policy perspective, this distinction is critical, as understanding the root causes of wage disparities informs the nature of policy implications and recommended mediations.

The economic dynamics of border cities are significantly shaped by disruptions in the global market, particularly due to the concentration of large enterprises and transnational corporations. This influence is especially notable given that foreign direct investment and exports either source from (Andrés-Rosales et al., 2019). According to Simmons et al, (2024), Gendered institutions—deeply ingrained traditions and practices within political, educational, healthcare, and economic systems—play a decisive role in shaping the opportunities and barriers faced by women. These institutional structures clearly evident in the persistent disparities in female labor force participation rates and in the wage gaps observed between men and women performing equivalent work across many nations.

According to Abraham, (2016), Empirical studies on formalization and wage inequality suggest that highly formalized compensation systems—those with clear, standardized rules—tend to mitigate sex-based discrimination. This achieved by limiting the discretionary power and subjective judgment of decision-makers during employee evaluations. Castellano and Rocca (2014) proposed the Gender Gap Labour Market Index (GGLMI), a novel composite indicator specifically focused on measuring gender disparities in labor markets, rather than general levels of development. This index designed to rank countries and contextualize these rankings within broader national economic, political, and social frameworks. It serves as a practical tool for policymakers by offering a clearer understanding of discrimination and inequality, thus guiding more targeted and effective policy decisions. A composite indicator, in this context, refers to a measure that reveals underlying trends or phenomena not immediately visible through individual indicators. It is particularly useful for presenting complex, multidimensional issues

in a simplified format that supports interpretation and decision-making.

Gender gap in the labor market is also challenge in Georgian Labor market. In order to encapsulate the gender gap in the labor market, various socioeconomic and statistical inferences must be drawn. This complex phenomenon involves numerous interrelated dimensions that must be carefully considered in both research and policymaking.

LITERATURE REVIEW

Empirical research of Cukrowska-Torzewska & Lovasz (2019) has consistently demonstrated that parenthood influences wages differently by gender. Specifically, having children tends to result in a decline in women's wages while slightly increasing men's wages. These patterns are commonly referred to as the "motherhood wage penalty" and the "fatherhood wage premium," respectively, and more broadly, as "family wage gaps." Parenthood shows to contribute significantly to the average wage differential between men and women, thereby influencing the overall gender wage gap.

Additional research of Calcagno & Montgomery (2020) identifies several other contributing factors to the wage gap. First, part of the disparity reflects differences in work experience and educational attainment between men and women. However, the gender gap in educational attainment has narrowed substantially and no longer serves as a primary explanatory factor. Differences in occupational preferences and negotiation behaviors also play a role. For instance, even within the same field or job role, women often work fewer hours, favoring occupations that offer greater scheduling flexibility. This preference is particularly evident among mothers, who are more likely than childless women to work reduced hours and consequently face a larger wage penalty.

Notably, women's publication rates in academic journals are increasing more rapidly than those of men, indicating a gradual reduction in the gender publication gap (GPG). Some journals—especially those in niche or micro-level domains—have already reached or are nearing gender parity in lead authorships. However, progress is slower in macro-level journals, with projections suggesting that parity may not be reached until 2045. These trends are linked to the increased representation of women in leadership positions and the implementation of targeted mentorship initiatives (Aguinis et al., 2024).

According to Simmons et al. (2024) women's early-stage entrepreneurial activity is positively correlated with national levels of gender equality. This relationship is amplified in institutional contexts where the social costs of failure are high, suggesting that societal

intolerance toward failure exacerbates the negative effects of gender inequality on women's participation in entrepreneurship.

To explore gender inequality and formalization in organizational contexts, recent scholarship begins by reviewing contemporary theories. According to Abraham (2016) on social identity theory, hypotheses are developed regarding the potential differences in how male and female managers utilize discretion under less formalized pay structures. The analytical framework assesses whether manager gender influences the extent to which informal pay practices contribute to gender-based wage disparities. Findings suggest that in certain cases, employees reporting to female managers' experience less inequality in unstructured compensation components compared to those reporting to male managers. These findings have important theoretical and managerial implications, particularly for understanding gender-based workplace inequality and for incorporating managerial attributes into future research on pay disparities.

While management intent was not directly assessed, social identity theory supports the assertion that male and female managers, when granted discretion through less formalized pay systems, may differ in their tendencies to favor men or disadvantage women. This theory posits that individuals categorize themselves and others based on observable demographic traits—such as gender—which can influence evaluation and decision-making processes (Abraham, 2016).

Historically, much of the research on gender wage inequality focused on mean wage differences. However, examining the mean alone provides an incomplete picture. Recent evidence, from both developed and developing economies, highlights variability in the wage gap across the wage distribution. According to Ahmed and Maitra, (2015), the "sticky floor" effect is evident at the lower end, where women earn significantly less than men are, while the "glass ceiling" effect persists at the upper end, where female workers also earn less than their male counterparts in higher-paying roles.

The factor price equalization theory holds that free trade tends to equalize the relative and absolute prices of production factors across countries. A key application of the Heckscher-Ohlin trade model is the Stolper-Samuelson theorem, which explains how changes in output prices affect input prices in a context of zero economic profit and positive production across industries. This framework is useful in analyzing the effects of trade liberalization or protectionist policies on factor incomes, under assumptions such as two countries, two goods, and two production factors operating under perfect competition (Almasifard, 2018).

From the human capital perspective, labor market

outcomes for men and women are seen as functions of the quality of their human capital—skills that should translate into productivity and, consequently, wages. Human capital theory argues that women's underperformance in the labor market results from rational choices that prioritize family over career, thus reducing their investment in education and job experience (Al-Waqfi & Al-Faki, 2015). In the previous research researchers mentioned statistical discrimination theory. Statistical discrimination theory further explains wage disparities by suggesting that employers use group membership—such as gender or ethnicity—as proxies for unobservable productivity traits. Employers, relying on historical averages, may make biased judgments about potential productivity based on gender, reinforcing inequality.

Cross-national research provides critical insights into gender dynamics in the labor force by viewing countries as distinct socio-institutional contexts. For instance, comparative analyses reveal how gender arrangements and institutional power dynamics shape women's labor market experiences. This research is particularly valuable in identifying mechanisms of pay setting and pinpointing discriminatory practices (Belgorodskiy et al., 2012). The study of Galbraith, (2019) have also shown that women typically have lower salary expectations than men, both during initial job negotiations and later when seeking raises. This expectation gap makes women more likely to accept lower offers and less likely to negotiate for salary increases. A proposed solution involves promoting wage transparency to correct expectations and encourage women to negotiate more effectively.

According to Georgiadis & Christopoulos, (2017), the most commonly cited explanation for the gender wage gap is discrimination—that is, wage disparities between workers with equal productive potential across different labor market segments. Another important factor is labor market segregation, as women tend to be overrepresented in lower-paying industries and occupations.

The Global Gender Gap Index provides a comprehensive tool for measuring and tracking gender-based disparities over time. It assesses national gaps in political empowerment, economic participation and opportunity, educational attainment, and health and survival. Crucially, the index evaluates the fairness of opportunity distribution, rather than linking outcomes to national income levels (Gokulsing & Tandrayen-Ragoobur, 2014). These insights highlight the limitations of Women in Development approaches and emphasize the need for a Gender and Development framework. The Gender and Development perspective shifts the focus to women's rights, participation, and agency, and critically ex-

amines how gendered power relations and social structures contribute to inequality.

According to Greer & Carden, (2020), human capital theory, wages reflect investments in general attributes such as age, education, job tenure, and informal training. Individuals with longer job tenures and higher educational attainment generally command higher wages. However, reports indicate that men, on average, possess greater educational qualifications and more job experience than women do. Moreover, organizational structures and job roles contribute to wage disparities, with women often concentrated in lower-paid, STEM-related positions ().

Roth, 2015 in the research mentioned gender stereotypes, males are come competent but not warm, and women are warm (kind, helpful, trustworthy, sympathetic, and altruistic) but not competent. In addition to prescribing how men and women should be, these stereotypes also set expectations about how they should be. Pay-for-performance based on performance standards that emphasize warmth may give women some evaluative advantage and close the gender gap, even though there is a hierarchy that generally places a higher value on skills related to competence than those related to warmth. Therefore, it is expected that Pay-for-performance based on culturally feminine performance criteria will reduce the gender gap, but Pay-for-performance based on culturally masculine performance criteria will increase it.

According to Almasifard (2018), Human capital constitutes the primary driver of a nation's prosperity. A country cannot progress unless its citizens are equipped with the necessary skills, knowledge, and understanding, and are capable of applying these assets to the national economy. In practice, innovative and competent individuals propel economic growth and development. Globally, men continue to participate in the labor force at significantly higher rates than women do. Regardless of whether they are self-employed or employed by others, working in businesses or agriculture, in paid service sectors, or performing unpaid domestic labor, women contribute substantially to national economies. Nonetheless, their contributions often fall short of their potential. Despite significant progress in recent decades, female labor force participation rates remain consistently lower than those of men do.

A major contribution of recent studies lies in illustrating how the labor market has contributed to increasing job instability. To ensure access to labor rights, regulatory frameworks have been established; however, these have often led to subcontracting and outsourcing practices. As a result, men are losing their wage advantage over women in specific productive sectors, particularly in regions characterized by high,

medium, and low exposure to global economic shifts (Andrés-Rosales et al., 2019). According to Calcagno & Montgomery (2020) key variables that explain gender wage disparities include marital status, professional experience, education level, and political tenure. However, disparities are more pronounced in Republican congressional offices compared to those of Independents or Democrats. Additionally, the Senate displays greater gender-based wage inequality than the House of Representatives, although significant disparities are also present within the House and among Democratic offices.

H1. There is positive linear relationship between Gender wage gap and sectors in Georgian labor market.

Several structural theories attempt to explain gender-based wage inequality, including social closure, occupational crowding, and the devaluation of female labor. Occupational crowding theory posits that wage levels are determined by the supply-demand balance within a profession. Women are often channeled into a limited number of “female-appropriate” jobs, creating labor surpluses in those fields and driving wages downward. Additionally, societal stereotypes devalue female-dominated work, leading employers to set lower wages even when men occupy the same roles. Empirical evidence supports both views, indicating that wages decline as the proportion of women in an occupation increases. Occupational and industrial segregation continue to be significant contributors to the gender wage gap (Fleming, 2015).

H2. There is positive linear relationship between Gender wage gap and in male-dominated sectors.

Women still frequently perform different types of work and are concentrated in different sectors compared to men, often under less favorable working conditions. These disparities, though varying by country, are prevalent in many parts of the world. To date, no country has fully closed the gender gap, making it a persistent global challenge (Al-Waqfi & Al-Faki, 2015). A key factor contributing to the slow pace of growth and the limited productivity structure in many sectors is the occupational separation of women into specific segments. Women are mainly employed in traditionally female-dominated occupations such as secretaries, nurses, teachers, waitresses, and unskilled service workers. This pattern of occupational gender segregation, much like wage discrimination, adversely affects women’s income levels and their opportunities for social mobility (Andrés-Rosales et al., 2019).

Educational attainment also plays a pivotal role in influencing female labor force participation. Beyond basic input factors, indicators of human capital reveal disparities in the returns to education, which reflect the numerous challenges and barriers that men and wom-

en encounter throughout the educational process (Castellano & Rocca, 2014). The labor market, as a critical component of the economy, significantly affects overall economic efficiency. In a dynamic and flexible labor market, workers can shift occupations relatively quickly, thereby encouraging employers to generate more job opportunities and increasing the employment-to-population ratio. Conversely, major obstacles to formal employment tend to push workers—especially those most vulnerable—into inactivity or low-productivity roles within the informal sector.

H3. There is positive linear relationship between Gender wage gap and in the gender population difference.

METHOD

The data used in this study is secondary and was obtained from the Georgian National Statistics Bureau. Specifically, the study employs panel data on average salaries across different sectors in Georgia, covering the period from 1999 to 2023. The main variables include the average monthly incomes of males and females. The dependent variable is the average salary, while the independent variables are sector, year, and gender-based population differences.

In addition, the dataset includes information on the employed population without social benefits, those receiving equal pay, and individuals without formal labor or union contracts. The data was obtained from **Geostat.ge** and is based on proportional distributions of the population with these characteristics.

Regression analysis was used to analyze the data collected. In this research, there is three hypothesis.

H1. There is positive linear relationship between Gender wage gap and sectors in Georgian labor market.

The **model summary** indicates a very strong fit, with an **R value of .966** and an **R² of .933**. This means that approximately **93.3% of the variance in average nominal salary** is explained by the predictors. The adjusted R² (.931) confirms the robustness of the model, with a relatively low standard error of estimate (134.39). The **ANOVA results** further demonstrate the overall significance of the model ($F = 329.852, p < .001$). This confirms that the predictors jointly contribute meaningfully to explaining variations in average salaries.

The **coefficients table** provides additional insights: The variable **Years** has a strong positive and highly significant effect on salaries ($B = 64.369, \beta = .919, t = 24.423, p < .001$). This implies that average monthly earnings have increased steadily over time. **Gender** also shows a positive and significant effect ($B = 302.238, \beta = .299, t = 7.951, p < .001$), suggesting that being male is associated with higher average monthly earnings com-

pared to being female. The **constant term** is negative (-129,232.347), reflecting the baseline adjustment of the regression equation, which is expected given the long time span of the data. Overall, the results confirm two key findings: **Gender disparities** remain statistically significant, with men consistently earning more than women.

Model Summary

Model	R	R Square	Adjusted R square	Std. Error of the Estimate
1	.966 ^a	.933	.931	134.39042

a. Predictors: (Constant), Years, Gender

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	11914757.759	2	5957378.880	329.852	.000 ^b
	Residual	848856.923	47	18060.786		
	Total	12763614.682	49			

a. Dependent Variable: AMENominal

b. Predictors: (Constant), Years, Gender

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-129232.347	5300.559		-24.381	.000
	Gender	302.238	38.011	.299	7.951	.000
	Years	64.369	2.636	.919	24.423	.000

a. Dependent Variable: AMENominal

H2. There is positive linear relationship between Gender wage gap and in male-dominated sectors.

The results of the ANOVA test reveal that sectors such as **Manufacturing** ($p = .032$), **Wholesale & Retail Trade** ($p = .040$), and **Finance & Insurance** ($p = .002$) show statistically significant differences across groups. Notably, these sectors are commonly classified as **male-dominated industries**, where male employment exceeds female employment by a substantial margin.

This finding lends partial support to **Hypothesis 2**, which posits a positive linear relationship between the gender wage gap and employment in male-dominated sectors. The presence of significant wage-related variation in these sectors may reflect the underlying structural inequalities that drive the gender wage gap, particularly where occupational segregation and industry norms favor higher wages for male-dominated roles.

However, further analysis is required to directly test the **linear relationship** implied in H2 — for example, using **regression analysis** to quantify how the degree of male dominance in a sector correlates with the size of the gender wage gap

H3. There is positive linear relationship between Gender wage gap and in the gender population difference.

The regression analysis demonstrates a **statistically significant relationship** between the population structure by age group and the gender wage gap ($R^2 = .833$, $p < .001$). Notably, the coefficient for the **45–65 age group** is both positive and significant ($B = 6.410$, $p = .001$), suggesting that a higher proportion of individuals in this older working-age cohort is associated with a wider gender wage gap. This result **supports Hypothesis 3 (H3)**, indicating that **differences in population composition — particularly within the labor force — influence the magnitude of gender-based wage disparities**. Interestingly, while the **0–19** group also shows a positive relationship ($B = 2.958$, $p = .034$), the **20–44** and **65+** groups display significant negative effects, which may reflect differences in labor force participation, retirement effects.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.913 ^a	.833	.818	217.49181

a. Predictors: (Constant), age groups(thousands)65+, age groups(thousands)20-44, age groups(thousands)0-19, age groups(thousands)45-65

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10634993.826	4	2658748.457	56.207	.000 ^b
	Residual	2128620.856	45	47302.686		
	Total	12763614.682	49			

a. Dependent Variable: AMENominal

b. Predictors: (Constant), age groups(thousands)65+, age groups(thousands)20-44, age groups(thousands)0-19, age groups(thousands)45-65

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3869.127	1034.498		3.740	.001
	age groups(thousands)0-19	2.958	1.352	.308	2.188	.034
	age groups(thousands)20-44	-9.878	1.089	-.917	-9.070	.000
	age groups(thousands)45-65	6.410	1.784	.567	3.592	.001
	age groups(thousands)65+	-3.728	.986	-.478	-3.780	.000

a. Dependent Variable: AMENominal

CONCLUSION

This study provides empirical evidence on the persistence and structural determinants of the gender wage gap in the Georgian labor market. The results demonstrate that gender remains a significant predictor of wage disparities, with men consistently earning higher average salaries across nearly all sectors. Regression and ANOVA analyses confirm strong statistical relationships between gender, sectoral composition, and demographic structure, indicating that both institutional and demographic factors continue to shape patterns of income inequality. In particular, male-dominated sectors such as manufacturing, wholesale and retail trade, and finance display notably wider wage differentials, reflecting deep-rooted occupational segregation and systemic barriers that limit women's economic advancement.

The findings further suggest that demographic composition—especially the prevalence of the 45–65 age group in the workforce—exacerbates the wage gap, underscoring the interaction between labor market dynamics and population characteristics. These results align with global research emphasizing the roles of human capital differentials, formalization levels, and discriminatory institutional practices in perpetuating gendered pay inequities.

From a policy perspective, the study highlights the urgent need for comprehensive gender-sensitive labor market reforms. Strengthening pay transparency, enforcing anti-discrimination frameworks, promoting women's participation in high-paying sectors, and implementing standardized compensation systems can help mitigate inequality. Future research should extend this analysis by incorporating micro-level organizational data and qualitative insights to better understand the mechanisms driving pay disparities. Ultimately, narrowing the gender wage gap is not merely a matter of fairness but a prerequisite for sustainable economic development and social equity in Georgia.

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